

Supplementary Data for Layered representation diagnostics for short metagenomic reads

Ruixiang Mei ^{1,*} Zhi Chen ² Rui Cao ³ Xunbing Gong ² and Teng Qi ²

¹School of Data Science, The Chinese University of Hong Kong, Shenzhen, 2001 Longxiang Road, Longgang District, Shenzhen 518172, Guangdong, China

²School of Medicine, The Chinese University of Hong Kong, Shenzhen, 2001 Longxiang Road, Longgang District, Shenzhen 518172, Guangdong, China

³School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen, 2001 Longxiang Road, Longgang District, Shenzhen 518172, Guangdong, China

*Correspondence: Ruixiang Mei, ruixiangmei@link.cuhk.edu.cn

Abstract

Supplementary figures and source tables supporting the representation-diagnostic analyses, including baseline, estimator, perturbation, parameter, redundancy and runtime audits.

Keywords supplementary data, metagenomics, representation diagnostics

Supplementary data overview

The canonical 4-mer, property and multi-scale property-pooling representation (CK4P-MSP) comprises a canonical 4-mer composition block (K), a global property block (P) and multi-scale property pooling (MSP). Figures S1–S7 report reduction, information-estimator, perturbation, counterfactual, parameter and dimension-matched baseline audits. Figures S8–S14 cover P/MSP relations, external stability, mechanism, historical descriptors, read-length continuity and fixed-head transfer. Figure S15 reports the supplementary CK4P-MSP-PKM positional-readout Pareto audit. Tables S1–S11 provide the corresponding dataset previews, contrasts, estimator checks, feature definitions, accessions, target metadata, coordinate-family attribution and unified runtime benchmark. Unless otherwise stated, plots show analysis-cell means without error bars. Cell-bootstrap intervals summarize prespecified matched cells that reuse source template panels and therefore do not represent independent biological cohorts.

Supplementary Methods: positional k-mer moment extension

CK4P-MSP-PKM was evaluated as a supplementary positional-moment extension of CK4P-MSP. For a read with $T = L - k + 1$ valid k -mer windows, the centre of window t was assigned a normalized relative coordinate $u_t \in [-1, 1]$. Let c_t denote the reverse-complement canonical 4-mer at that window and let

$$\mathbf{b}(u_t) = \left[u_t, \frac{3u_t^2 - 1}{2}, \frac{5u_t^3 - 3u_t}{2} \right]$$

be the first three non-constant Legendre-style position basis functions. The uncompressed positional k -mer moment array was

$$R_i(c, q) = \sum_{t=1}^T \mathbb{1}(c_t = c) b_q(u_t),$$

where c indexes the 136 canonical 4-mers and $q \in \{1, 2, 3\}$. A deterministic signed hash with seed 20260805 mapped the 408 token-moment coordinates to a 75-dimensional vector, which was row-normalized to give $\hat{\mathbf{Z}}_i$. Invalid windows containing ambiguous bases contributed zero to the moment array.

The supplementary extension retained the normalized CK4P-MSP blocks and appended PKM as

$$\mathbf{x}_i^{\text{PKM}} = \frac{1}{\sqrt{3 + \delta^2}} \left[\hat{\mathbf{K}}_i; \hat{\mathbf{P}}_i; \hat{\mathbf{M}}_i; \delta \hat{\mathbf{Z}}_i \right].$$

The fixed grid $\delta \in \{0.1, 0.25, 0.5, 0.75, 1\}$ was declared before the exploratory sweep. Among the tested weights, $\delta = 0.25$ met the prespecified screen for motif-position gain, bounded WGS/ART drift change, retained nearest-clean retrieval, local-change readout, CAMI fixed-head retention and less than two-fold reference extraction time. It is reported as the exploratory CK4P-MSP-PKM Pareto setting.

The sweep reused the WGS, local-change, public Dorsal motif-position, ART and CAMI resources defined in the main Methods. The motif-position probe comprised 1,125 windows from 32 cis-regulatory modules, and all windows from one module were retained in the same cross-validation fold. A seed-20260819 resampling check quantified seed sensitivity within the same benchmark resources. Extraction cost was taken from the unified long-duration runtime benchmark, and the direction audit used 5,000 clean/reverse-complement pairs. Because odd position moments reverse sign, CK4P-MSP-PKM is input-order-sensitive; only its CK4 block is canonicalized.

Historical descriptors were evaluated under the same sampled protocol to position the extension within the observed stability–readout–cost trade-offs.

Supplementary Figures

Fitted projection can improve drift; CK4P-MSP remains training-free and block-decomposable

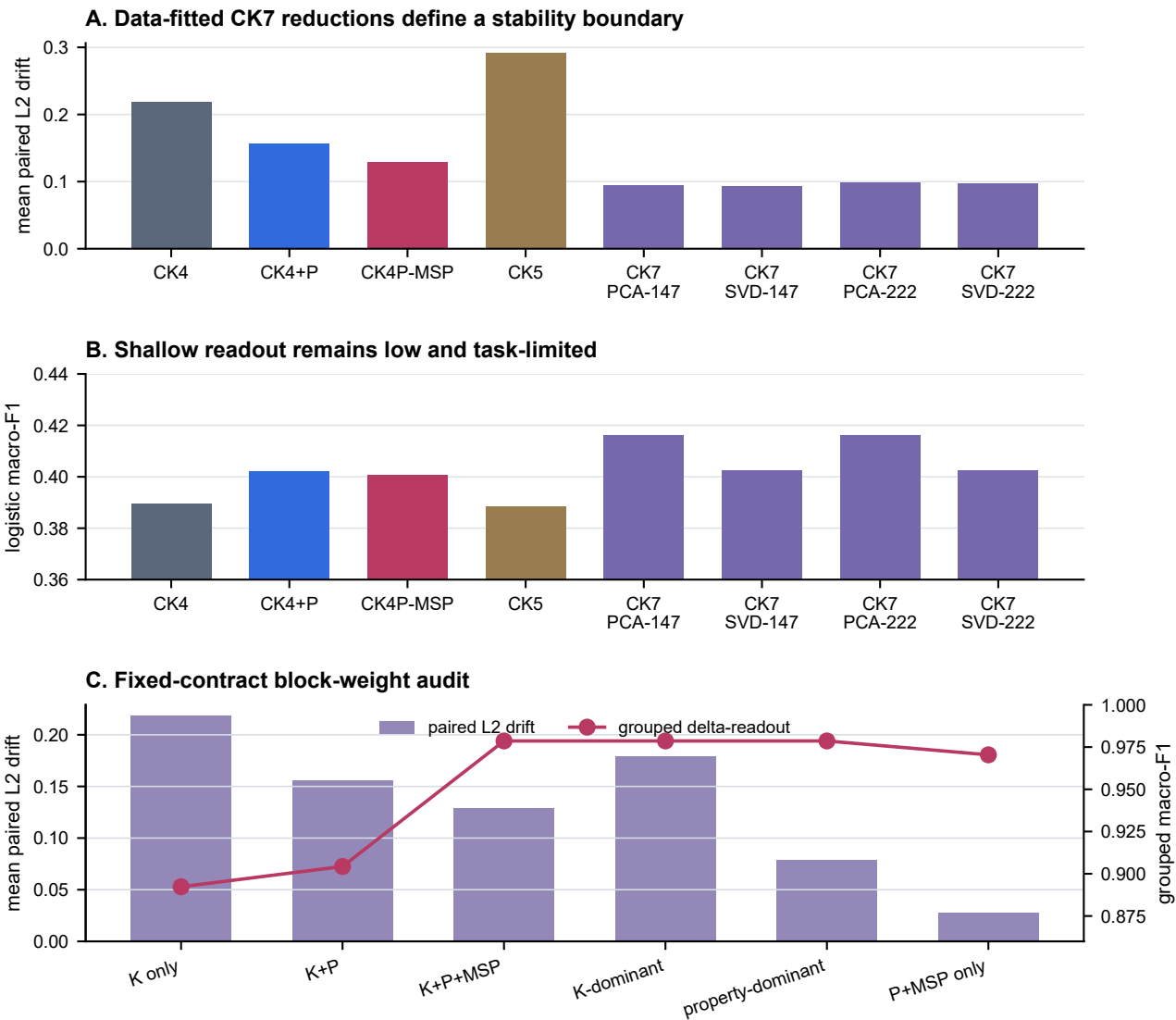
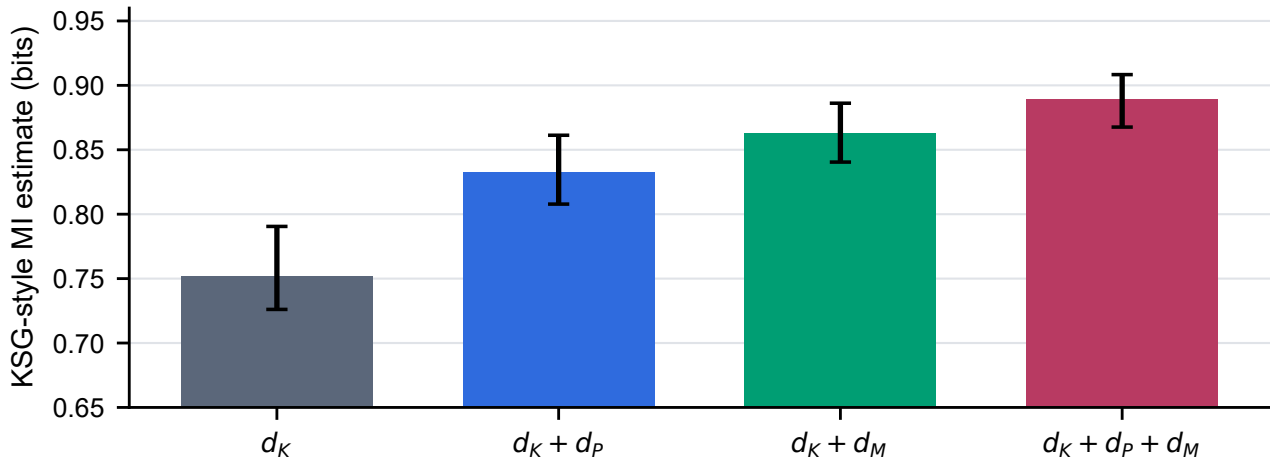


Figure S1 Fitted-reduction and block-weight boundary audit. (A) Mean standardized representation drift across 24 controlled WGS length-by-perturbation cells for CK4/CK5, CK4+P, CK4P-MSP and clean-partition-fitted CK7 PCA/SVD projections at 147 or 222 target dimensions; lower is preferable. The fitted CK7 projections achieve lower drift than CK4P-MSP and therefore define a data-dependent stability boundary. (B) Mean logistic macro-F1 across 132 WGS readout cells; projection rank in these smaller per-cell training partitions was sample-limited (median 49.5 components), so this panel is a shallow-readout boundary rather than a strict 147/222-dimensional comparison. (C) Mean drift across 24 WGS cells and grouped local-change macro-F1 across 12 cells under K-only, K+P, K+P+MSP and prespecified K/property-dominant weights. Bars, points and lines are cell means without uncertainty intervals. Grouped readout was unchanged across the three non-zero K/P/MSP weightings, indicating low sensitivity to the tested weights.

Alt text: Three panels show that fitted CK7 PCA/SVD projections have lower drift than CK4P-MSP, all shallow readouts remain low, and the grouped local-change result is stable across non-zero K/P/MSP weightings.

S2. Estimator-dependent local-versus-noise separability audit



Error bars: 2.5th-97.5th percentiles across stratified subsamples.
Signed joint-minus-K increment = 0.138 bits; cell bootstrap 95% CI 0.073-0.207.
Aggregate label-permutation P = 0.0099.

Figure S2 Estimator-dependent mutual-information (MI) audit for local change versus matched nuisance perturbation across 12 length-by-local-mode cells. Bars show KSG-style estimates for d_K , $d_K + d_P$, $d_K + d_M$ and $d_K + d_P + d_M$; error bars are the 2.5th–97.5th percentiles across 100 stratified subsamples per cell. The signed joint-minus-K increment was evaluated with a 10,000-resample paired-cell bootstrap and 100 shared label permutations per cell. Values quantify estimator-dependent local-versus-noise separability within this grid; higher values indicate greater separability.

Alt text: Bar plots summarizing empirical mutual-information and conditional-mutual-information audit values across representation blocks.

ART error load varies by cycle profile; property-aware stability persists within strata

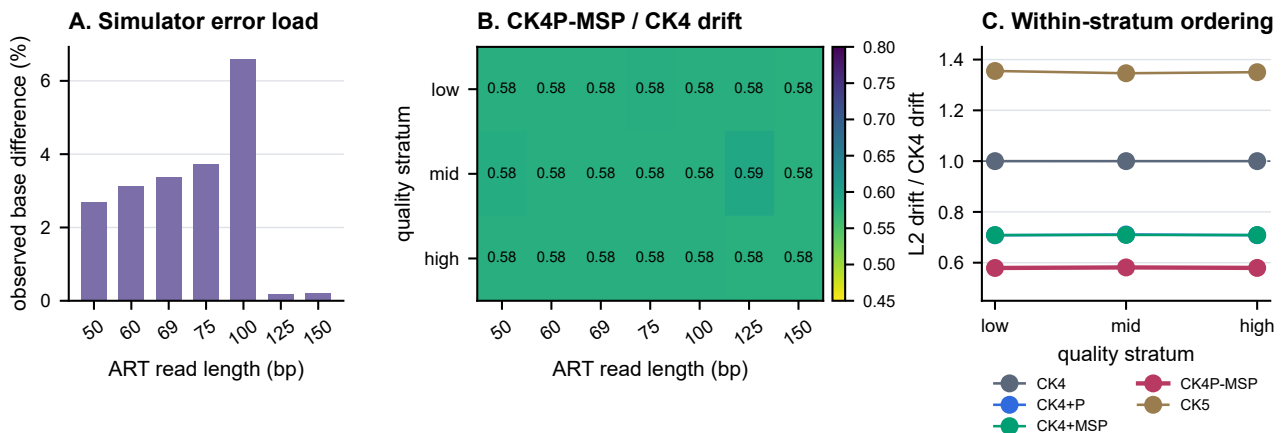


Figure S3 ART error-load and quality-stratification audit using 55,961 matched clean/simulated pairs, represented as 111,922 rows, at 50, 60, 69, 75, 100, 125 and 150 bp. (A) Bars show the observed same-coordinate base-difference rate at each length; this descriptive mismatch rate is not an alignment-aware edit distance. (B) Heatmap cells show CK4P-MSP standardized L2 drift divided by CK4 drift within each length and low-, middle- or high-quality stratum; values below one indicate less drift than CK4. (C) Lines show mean within-stratum drift ratios for CK4, CK4+P, CK4+MSP, CK4P-MSP and CK5, averaged over the seven lengths. Panels report descriptive means without uncertainty intervals. The selected ART cycle profile produces a non-monotonic error load across lengths, so comparisons are interpreted within length and quality stratum.

Alt text: Three panels show non-monotonic ART base-difference rates, CK4P-MSP-to-CK4 drift ratios by length and quality, and stable within-stratum representation ordering.

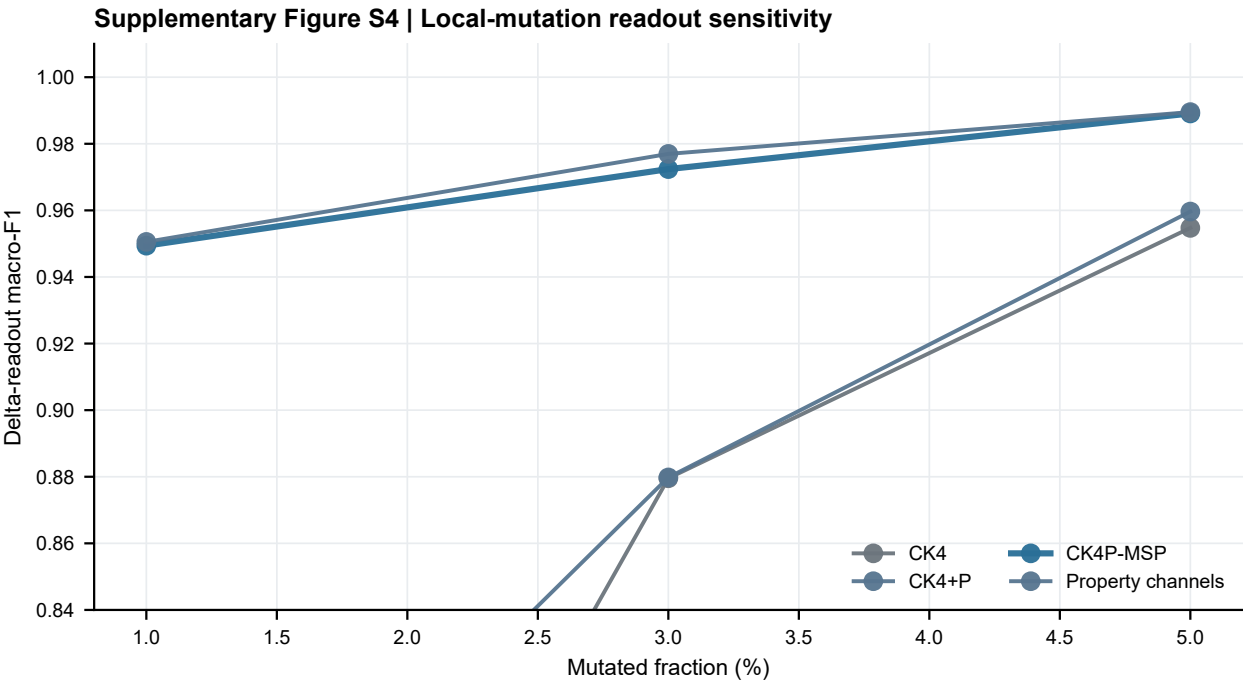


Figure S4 Local-change fraction sweep at 1%, 3% and 5% changed positions. Curves show cell-mean logistic grouped delta-readout macro-F1 without uncertainty intervals, averaged across 12 length-by-local-mode cells with 180 template triplets per cell and all derivatives of one template retained in one fold. Higher values indicate greater accessibility of local-versus-nuisance structure. CK4+MSP and CK4P-MSP retain the strongest readout across effect sizes. The endpoint is synthetic local-change accessibility.

Alt text: Line plot showing local mutation-fraction sensitivity across compact representation families.

Supplementary Tables

Table S1. Preview of the current CAMI II marine anonymous-read source table. The 12 rows show the four prespecified compact representations at 69, 75 and 100 bp under 3% N masking; each row summarizes 4,000 clean/perturbed pairs. The complete length-by-perturbation CSV is retained in the data package.

length	condition	method	features	paired cosine	L2 drift	top-1
69	N_3pct	CK4	136	0.9643	0.2649	1.0000
69	N_3pct	CK4+P	147	0.9812	0.1926	1.0000
69	N_3pct	CK4P-MSP	222	0.9871	0.1594	1.0000
69	N_3pct	CK5	512	0.9383	0.3487	1.0000
75	N_3pct	CK4	136	0.9683	0.2497	0.9998
75	N_3pct	CK4+P	147	0.9833	0.1814	0.9998
75	N_3pct	CK4P-MSP	222	0.9886	0.1501	1.0000
75	N_3pct	CK5	512	0.9440	0.3324	1.0000
100	N_3pct	CK4	136	0.9686	0.2486	1.0000
100	N_3pct	CK4+P	147	0.9833	0.1816	1.0000
100	N_3pct	CK4P-MSP	222	0.9886	0.1500	1.0000
100	N_3pct	CK5	512	0.9406	0.3428	1.0000

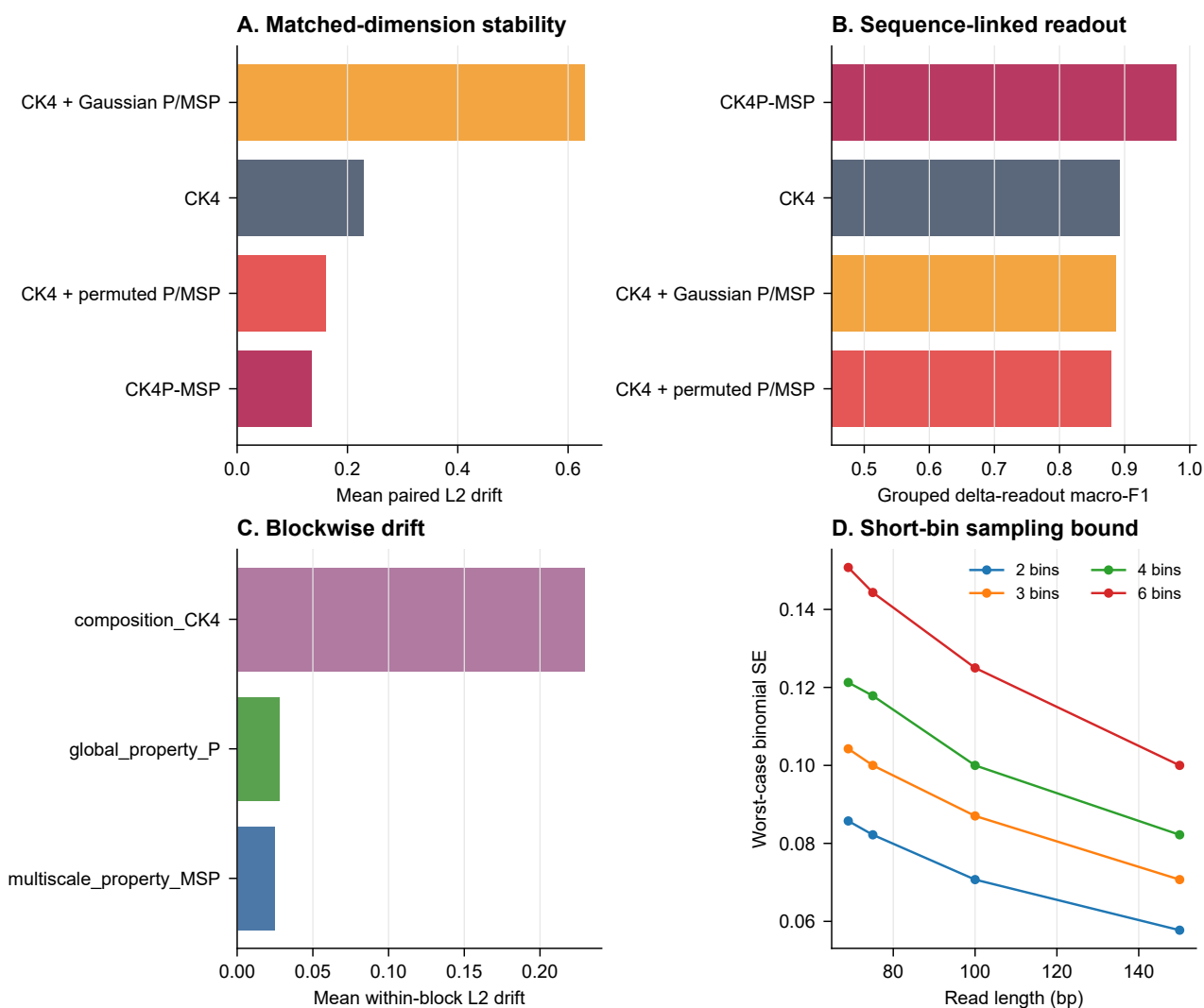


Figure S5 Sequence-linked P/MSP counterfactual and short-bin reliability audit. (A) Mean standardized drift across 20 WGS length-by-perturbation cells compares CK4P-MSP with CK4 and with row-permuted or marginally matched Gaussian P/MSP controls under the fixed block-normalization scheme. (B) Mean logistic grouped delta-readout across 12 local-change cells (250 template triplets per cell) is higher for sequence-linked CK4P-MSP (0.979) than for CK4 (0.892), permuted P/MSP (0.879) or Gaussian P/MSP (0.886). (C) Cell-mean K, P and MSP blockwise drift separates their perturbation responses. Panels A–C do not show uncertainty intervals. (D) Worst-case binomial standard error is shown for 2-, 3-, 4- and 6-bin scales over the tested WGS anchor lengths, based on the shortest bin at each scale. Lower drift and sampling error and higher readout are preferable on their respective panels.

Alt text: Multi-panel audit of P-channel counterfactual behaviour, drift decomposition and short-bin sampling bounds.

Static MSP sensitivity across representative WGS conditions and the 50 to 75 bp sweep

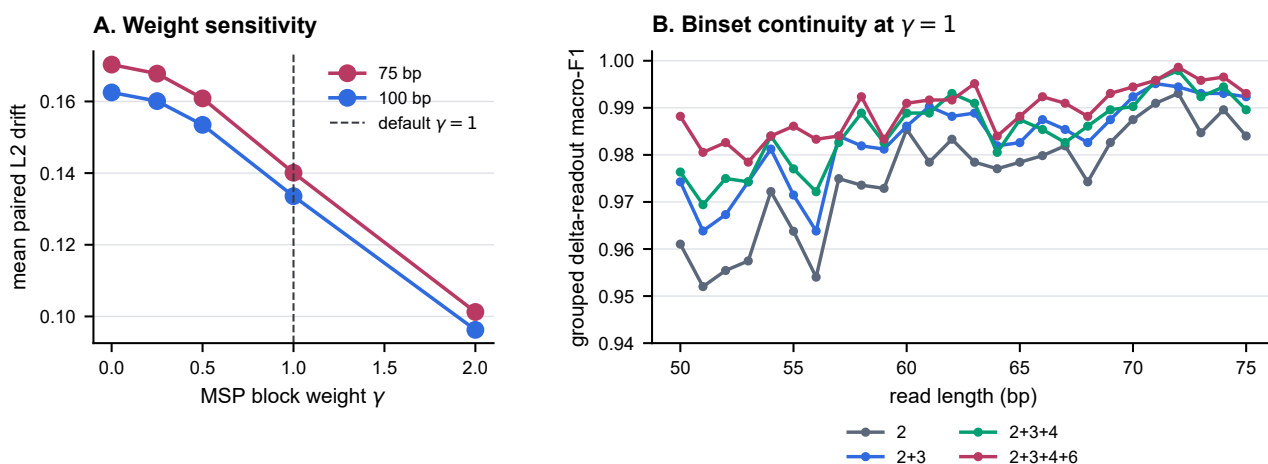


Figure S6 MSP binset and weight-sensitivity audit. (A) Points are mean drift without uncertainty intervals over the prespecified WGS perturbations at 75 and 100 bp for the complete 2+3+4+6 binset and $\gamma \in \{0, 0.25, 0.5, 1, 2\}$; lower is preferable. (B) Lines are mean grouped delta-readout macro-F1 over four local modes at every integer length from 50 to 75 bp, using 180 template triplets per length-by-mode cell at $\gamma = 1$ for cumulative binsets 2, 2+3, 2+3+4 and 2+3+4+6; higher is preferable. All derivatives of one template were retained in the same fold during fivefold grouped cross-validation. No uncertainty intervals are shown, and no parameter selection was performed from this audit.

Alt text: Line plots show gamma-weight sensitivity at 75 and 100 bp and MSP binset continuity across every length from 50 to 75 bp.

High-k vector controls and MinHash require separate geometries

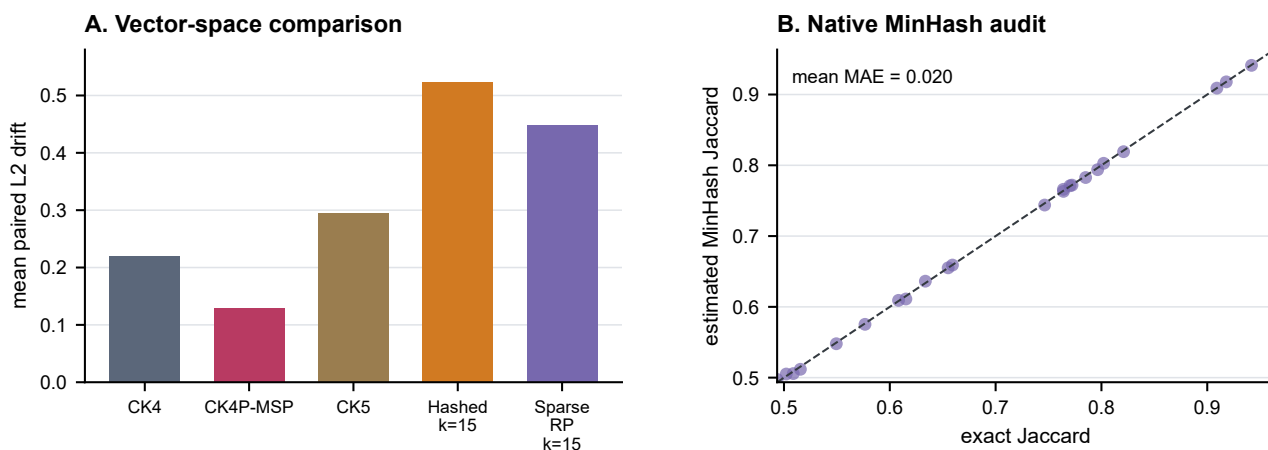


Figure S7 Dimension-matched high-k and native MinHash audit across 24 controlled WGS length-by-perturbation cells. (A) Bars are cell-mean standardized representation drift without uncertainty intervals for hashed and sparse-random-projection canonical $k = 15$ vectors, CK4, CK5 and CK4P-MSP at a 222-feature budget; lower is preferable. (B) Each point compares 222-hash MinHash signature agreement with exact $k = 15$ Jaccard similarity in one matched cell; the diagonal denotes equality and lower absolute error indicates better approximation. MinHash is not treated as an L2 feature vector.

Alt text: Multi-panel audit comparing dimension-matched high-k compressed vectors and native MinHash agreement with exact Jaccard similarity.

P/MSP contribution and relation audit

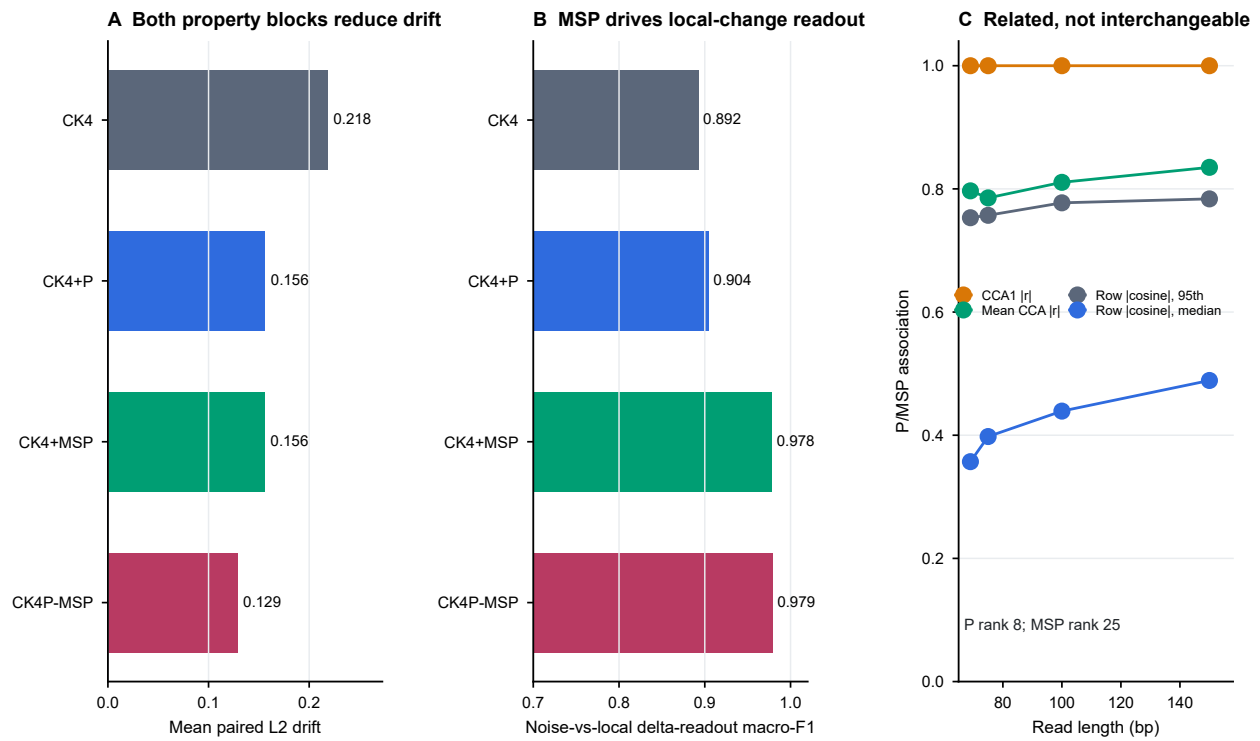


Figure S8 P/MSP contribution and relation audit. (A) Bars are mean paired L2 drift without uncertainty intervals across 24 controlled WGS length-by-perturbation cells for CK4, CK4+P, CK4+MSP and CK4P-MSP; lower is preferable. (B) Bars are mean template-grouped local-versus-nuisance delta-readout macro-F1 without uncertainty intervals across 12 length-by-local-mode cells with 250 template triplets per cell; higher is preferable. (C) Descriptive canonical correlations and row-wise alignment are calculated from 600 clean reads at each of 69, 75, 100 and 150 bp; no inferential intervals are shown. P and MSP share a strong leading nucleotide-property component, whereas incomplete row-wise alignment and different numerical ranks support related-but-non-equivalent wording.

Alt text: Three panels show that P and MSP both reduce paired drift, MSP contributes most of the grouped local-change readout gain, and the two property blocks share leading structure without being row-wise interchangeable.

Table S2. Prespecified conditional-contribution contrasts from the seven-group K/P/MSP ablation. The full representation is CK4P-MSP. Improvement is signed so that positive values favour the full representation; for L2 drift, the sign is reversed because lower values are preferable. Confidence intervals are percentile intervals from 10,000 paired-cell bootstrap resamples. Two-sided Wilcoxon signed-rank tests were adjusted across all 12 rows by the Benjamini–Hochberg procedure. Global cells are length-by-perturbation pairs; local cells are length-by-local-mode grouped cross-validation summaries. Cells reuse source template panels, so the intervals and tests quantify consistency across the prespecified grid rather than independent biological-cohort inference.

block contrast	evidence layer	metric	<i>n</i>	full	comparator	improvement [95% CI]; BH <i>q</i>
K P+MSP	global	paired cosine	24	0.989	0.999	−0.010 [−0.013, −0.008]; 2.4×10^{-7}
K P+MSP	global	L2 drift	24	0.129	0.027	−0.101 [−0.118, −0.086]; 2.4×10^{-7}
K P+MSP	global	retrieval	24	1.000	0.942	0.058 [0.036, 0.081]; 7.3×10^{-4}
K P+MSP	local	grouped macro-F1	12	0.979	0.970	0.008 [−0.0002, 0.019]; 0.507
P K+MSP	global	paired cosine	24	0.989	0.984	0.005 [0.004, 0.006]; 2.4×10^{-7}
P K+MSP	global	L2 drift	24	0.129	0.156	0.027 [0.023, 0.031]; 2.4×10^{-7}
P K+MSP	global	retrieval	24	1.000	1.000	0.000 [0.000, 0.000]; 1.000
P K+MSP	local	grouped macro-F1	12	0.979	0.978	0.001 [−0.0002, 0.002]; 0.525
MSP K+P	global	paired cosine	24	0.989	0.984	0.005 [0.004, 0.006]; 2.4×10^{-7}
MSP K+P	global	L2 drift	24	0.129	0.156	0.027 [0.023, 0.032]; 2.4×10^{-7}
MSP K+P	global	retrieval	24	1.000	1.000	0.000 [0.000, 0.000]; 1.000
MSP K+P	local	grouped macro-F1	12	0.979	0.904	0.074 [0.035, 0.117]; 7.3×10^{-4}

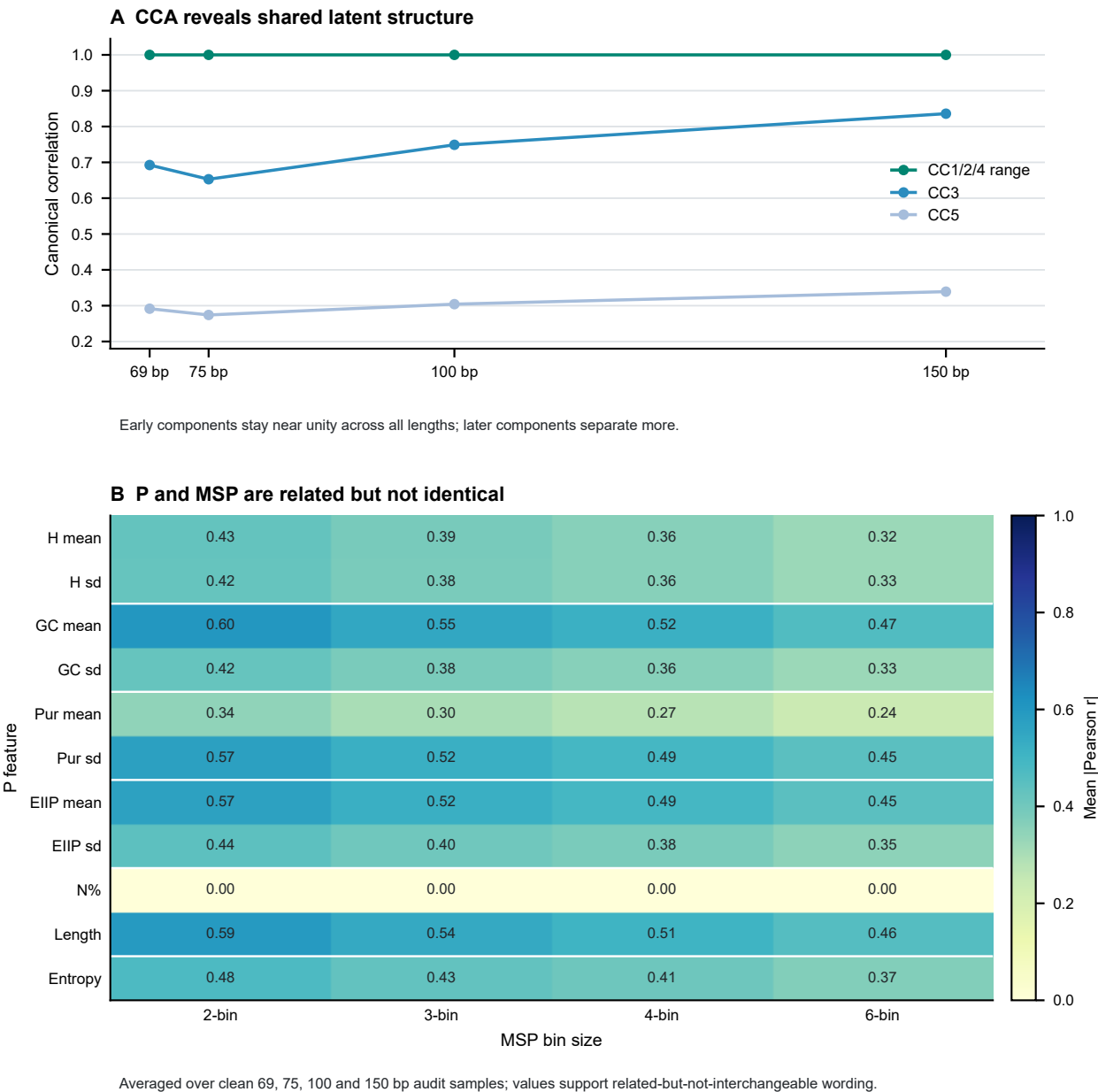


Figure S9 P/MSP relation audit using 600 clean reads at each of 69, 75, 100 and 150 bp. (A) Canonical-correlation analysis summarizes shared latent structure across the 11-dimensional P and 75-dimensional MSP blocks. (B) The heatmap reports mean absolute Pearson correlation between each P coordinate and each MSP bin-scale family, averaged across the four lengths. Values are descriptive estimates without uncertainty intervals. P and MSP share a leading latent component but retain distinct rank and row-level structure.

Alt text: Line and heatmap panels showing CCA and correlation relationships between P and MSP feature layers.

Table S3. Estimator-dependent empirical MI separability increment across 12 prespecified length-by-local-mode cells. The contrast is signed and was not clipped at zero. The confidence interval uses 10,000 paired-cell bootstrap resamples; the aggregate permutation *P* value uses 100 shared label permutations per cell. The interval describes analysis-grid variation and is not an independent biological-cohort confidence interval.

contrast	<i>n</i> cells	mean (bits)	95% CI	Wilcoxon <i>P</i>	permutation <i>P</i>	<i>k</i>
$\hat{I}(d_K, d_P, d_M; Y) - \hat{I}(d_K; Y)$	12	0.138	[0.073, 0.207]	0.0004883	0.009901	5

CAMI II marine anonymous-read stability probe
n=4,000 reads; 48,000 length/condition rows; unlabeled perturbation-stability probe

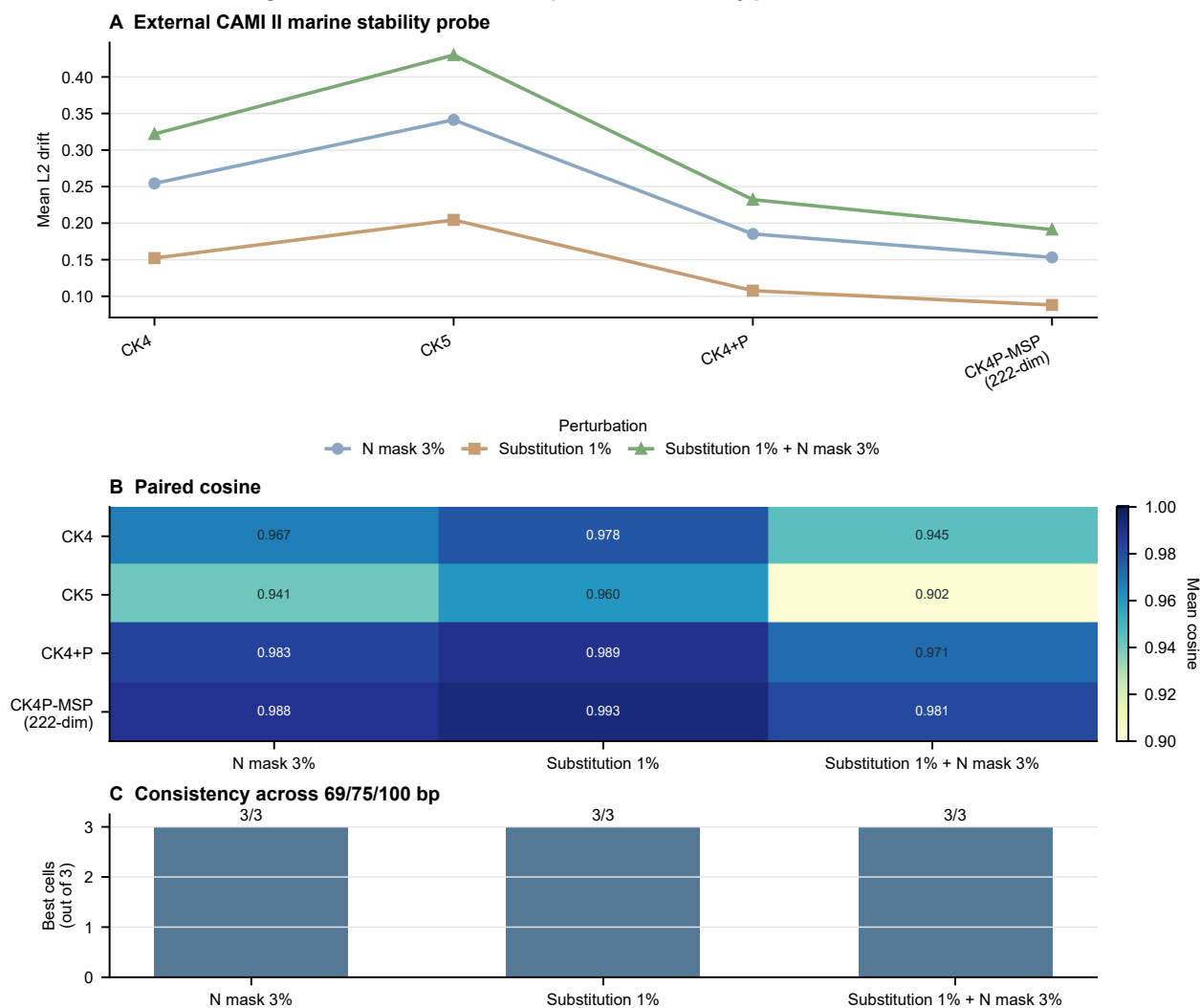


Figure S10 CAMI II marine anonymous-read stability probe using 4,000 anonymous source reads, expanded to 48,000 length-by-condition rows after truncation to 69, 75 and 100 bp and application of N masking, substitution and combined perturbations. (A) Lines show mean standardized L2 drift across the three lengths for each representation and perturbation; lower is preferable. (B) Heatmap cells show mean paired cosine across the three lengths; higher is preferable. (C) Bars count, for each perturbation, the number of length cells out of three in which CK4P-MSP had the lowest L2 drift among the displayed methods. Panels show descriptive cell means or counts without uncertainty intervals. Read-level taxonomic labels were not reconstructed, so this figure supports external stability only.

Alt text: Line, heatmap and bar panels showing CAMI II marine anonymous-read stability across length and perturbation settings.

MSP exposes spatial structure; map rescaling defines a boundary

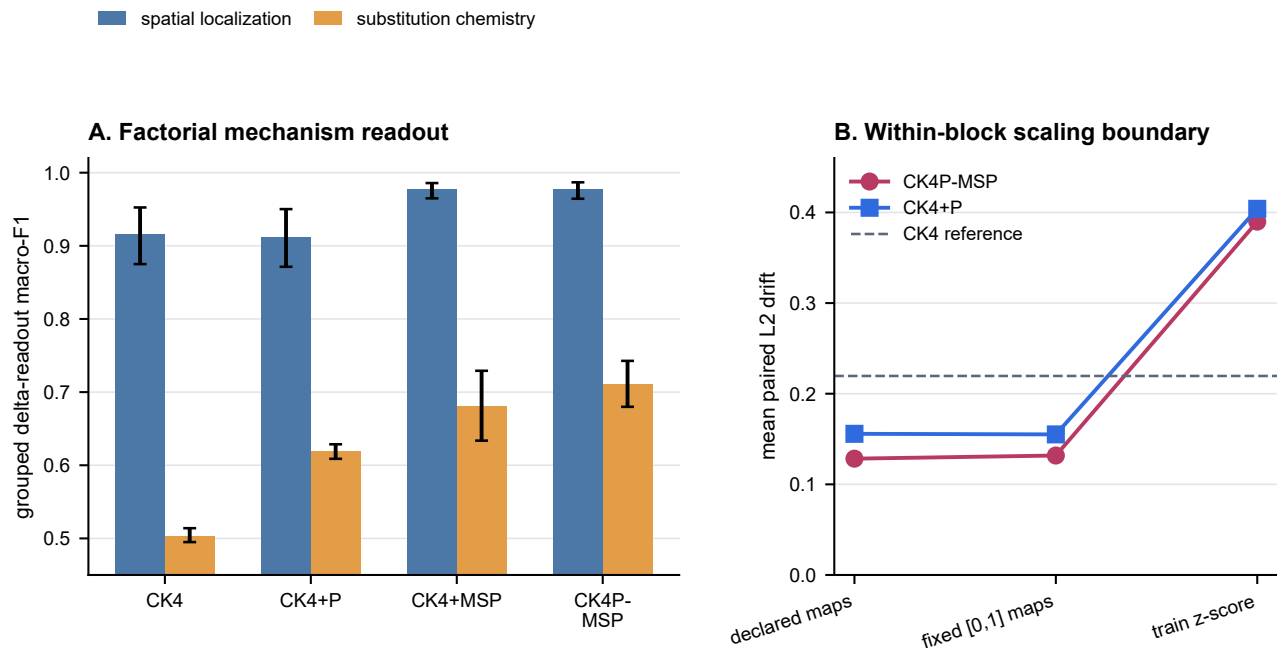


Figure S11 Mechanism-aligned local-change and property-scaling audits. (A) Grouped macro-F1 for spatial localization (contiguous versus dispersed substitutions) and substitution chemistry (property-directed versus random substitutions) across the four K-containing representations. Both tasks used 3,000 template-level observations per representation across 12 length-by-local-mode cells, with all derivatives of a template retained in one fold; error bars are 95% bootstrap intervals across cells. (B) Mean standardized representation drift across 24 WGS length-by-perturbation cells under the declared mappings, fixed per-property [0, 1] mappings and train-fit coordinate z-scores. Higher macro-F1 and lower drift are preferable; train-fit z-scoring is shown as a preprocessing sensitivity analysis.

Alt text: Two panels showing that MSP improves spatial and chemistry readout, P adds a smaller chemistry increment, and fixed property remapping preserves drift while train-fit z-scoring does not.

Table S4. Prespecified conditional block contrasts in the 2 × 2 spatial-localization-by-substitution-chemistry audit. Macro-F1 was estimated by template-grouped fivefold cross-validation. Confidence intervals are 10,000-resample paired-cell bootstrap intervals, and *q* values are Benjamini–Hochberg adjusted across the six contrasts. The matched cells share the controlled template panel; inference is therefore restricted to consistency across this experimental grid.

conditional block	readout target	full	comparator	Δ macro-F1 [95% CI]	BH <i>q</i>
K P+MSP	spatial localization	0.977	0.964	0.013 [0.001, 0.025]	0.1553
K P+MSP	substitution chemistry	0.710	0.758	-0.048 [-0.058, -0.037]	0.0009766
P CK4+MSP	spatial localization	0.977	0.977	0.000 [-0.001, 0.002]	0.5195
P CK4+MSP	substitution chemistry	0.710	0.680	0.030 [0.013, 0.048]	0.03149
MSP CK4+P	spatial localization	0.977	0.912	0.065 [0.034, 0.098]	0.0009766
MSP CK4+P	substitution chemistry	0.710	0.618	0.092 [0.059, 0.130]	0.0009766

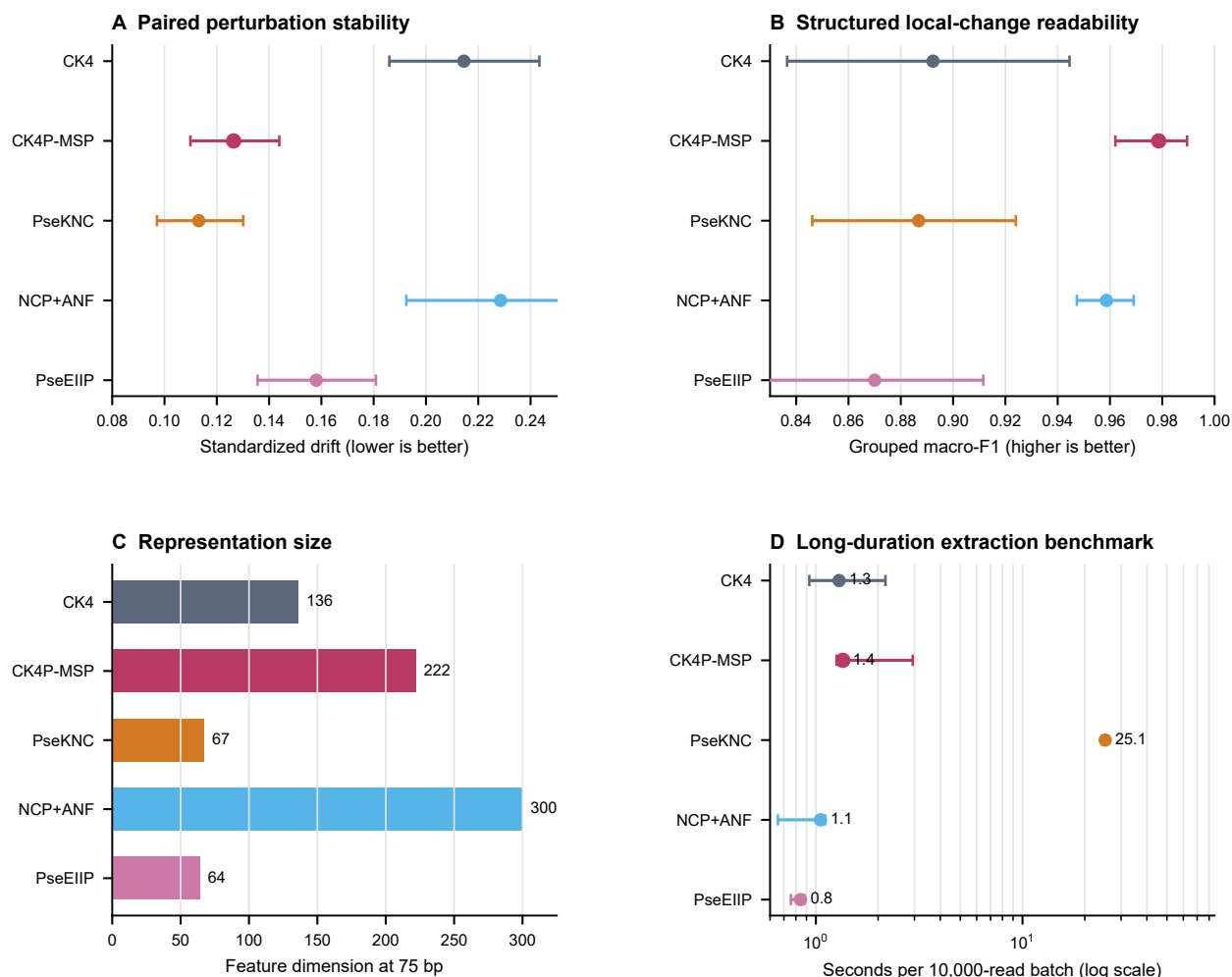


Figure S12 Historical handcrafted descriptor boundary audit. (A) Mean standardized representation drift with 95% matched-cell bootstrap intervals across 30 WGS length-by-perturbation cells; lower is preferable. PseKNC has lower drift than CK4P-MSP. (B) Logistic grouped local-versus-nuisance delta-readout macro-F1 with 95% matched-cell bootstrap intervals across 12 length-by-local-mode cells; all derivatives of a source template remained in one fold, and higher is preferable. CK4P-MSP has the strongest grouped local-change readout in this comparison. These intervals quantify variation across the sampled analysis grid; independent cohorts were not sampled. (C) Feature dimensions at 75 bp. NCP+ANF scales with the declared read-length budget, whereas the other descriptors are fixed width. (D) Full-run median wall-clock extraction time and interquartile range from the unified 10,000-read, 75-bp benchmark, shown on a logarithmic axis. Each method accumulated at least 120 timed seconds and five calls after a 100-read warm-up. Methods ran in randomized round-robin order, garbage collection preceded each call and numerical libraries were limited to one thread. A separate 100,000-read pass assessed scaling. Timings were obtained on an Intel Core Ultra 5 125H workstation with 31.6 GB RAM and are implementation-specific. PseKNC used $k = 3$, $\lambda = 3$ and $w = 0.05$; deterministic ambiguous-base extensions are defined in Methods. The panels expose distinct stability, readout, dimension and runtime trade-offs.

Alt text: Four panels show that PseKNC has the lowest drift and dimension, CK4P-MSP has the highest grouped local-change readout and near-CK4 optimized runtime, and NCP+ANF is position-explicit but length-dependent in dimension.

CK4P-MSP retains its stability-readout profile across 50 to 75 bp

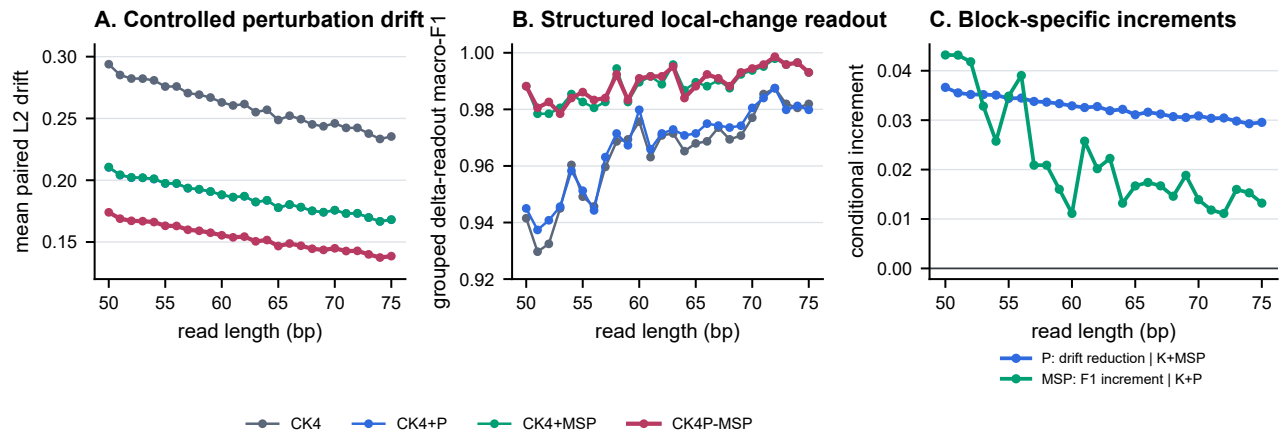


Figure S13 Shared-template continuity audit across every integer length from 50 to 75 bp. (A) Mean paired standardized L2 drift over six controlled perturbations for CK4, CK4+P, CK4+MSP and CK4P-MSP; lower is preferable. Each point summarizes 420 shared source templates that were prefix-truncated from the same 150-bp windows. (B) Mean grouped local-versus-nuisance delta-readout macro-F1 over four local modes with 180 template triplets per length-by-mode cell and fivefold template-grouped cross-validation; higher is preferable. (C) Conditional block-specific increments: P-associated drift reduction is CK4+MSP minus CK4P-MSP, whereas the MSP-associated readout increment is CK4P-MSP minus CK4+P. Positive values indicate improvement on the stated metric. Lines show descriptive analysis-cell means without uncertainty intervals and isolate lower-range length sensitivity.

Alt text: Three line panels show smooth drift reduction, high grouped local-change readout and positive metric-specific P and MSP increments from 50 to 75 bp.

Frozen-head audits distinguish representation geometry from learned preprocessing

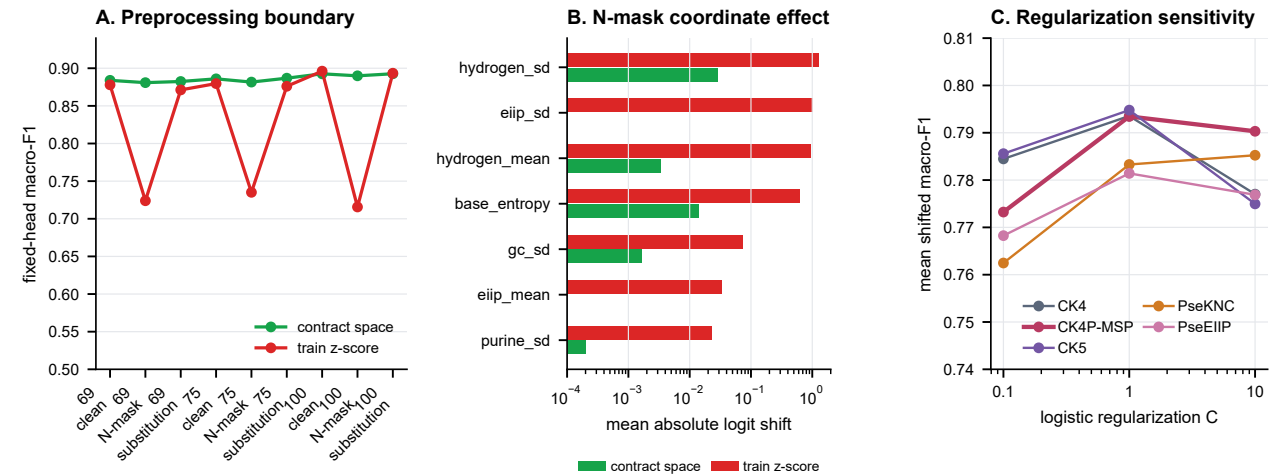


Figure S14 Frozen-head preprocessing and regularization sensitivity. (A) Macro-F1 for CK4P-MSP in a separate single-target preprocessing-sensitivity audit, distinct from the primary six-target cross-target summary in Supplementary Table S8. Block-normalized probes use the declared representation directly; train z-score probes apply coordinate-wise means and variances fitted only on the 100-bp clean training fold. Each point aggregates 2,486 held-out source groups through fivefold source-grouped cross-validation. (B) Mean absolute logistic-logit shifts of the seven most affected P coordinates under 100-bp, 3% N masking, averaged over five folds. Green bars denote block-normalized contributions and red bars denote amplification after train-fit z-scoring; the horizontal axis is logarithmic. (C) Mean shifted macro-F1 across six CAMI target tasks and eight non-baseline length-by-condition cells per target at $C = 0.1, 1$ and 10 . The classifier was otherwise unchanged and was never refitted on shifted conditions. $C = 1$ was prespecified; the other values quantify regularization sensitivity. These panels show that low representation drift does not guarantee invariance to learned downstream preprocessing.

Alt text: A line panel shows N-mask-specific degradation after train-fit z-scoring, a log-scale bar panel identifies amplified low-variance nucleotide-property coordinates, and a final line panel shows regularization-dependent fixed-head readout.

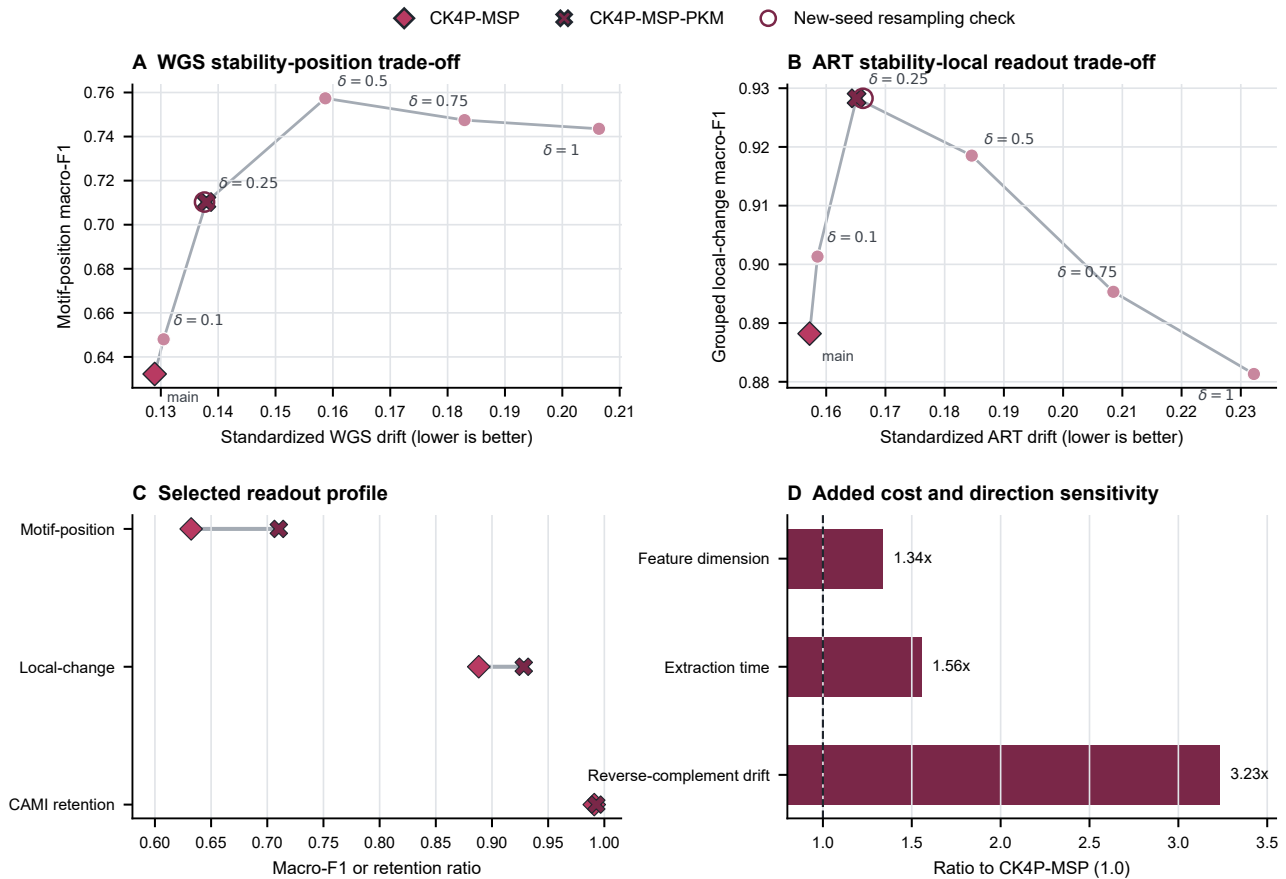


Figure S15 Exploratory CK4P-MSP-PKM positional-readout Pareto audit. (A) The prespecified PKM weight path compares mean WGS standardized drift across 24 length-by-perturbation cells with motif-position macro-F1; lower drift and higher readout are preferable. (B) The same path compares mean ART drift across seven read lengths with the exploratory candidate-route local-readout macro-F1; lower drift and higher readout are preferable. The local-readout values in panels B and C are from this candidate screen and are not the common historical-descriptor estimate reported in the main text. Diamonds mark CK4P-MSP, crosses mark the selected CK4P-MSP-PKM point at $\delta = 0.25$, and open circles mark the new-seed resampling check within the same benchmark resources. (C) Dumbbells compare the two methods on motif-position macro-F1, candidate-route local-readout macro-F1 and CAMI fixed-head retention. (D) Bars report the CK4P-MSP-PKM-to-CK4P-MSP ratios for feature dimension, full-run median extraction time from the unified long-duration benchmark, and mean reverse-complement-pair drift over 5,000 pairs. Values above one indicate added cost or direction sensitivity. CK4P-MSP-PKM defines a higher-readout, higher-cost Pareto point.

Alt text: Four panels show the fixed PKM weight trade-off, selected structured-readout gains and the corresponding dimension, runtime and reverse-complement direction-sensitivity costs relative to CK4P-MSP.

Table S5. Property-scaling sensitivity across 24 WGS length-by-perturbation cells. Fixed $[0, 1]$ maps and train-fit coordinate z-scores test sensitivity relative to the declared primary mapping.

representation	property scaling	paired cosine	L2 drift	retrieval
CK4P-MSP	declared maps	0.989	0.128	1.000
CK4P-MSP	fixed $[0,1]$ maps	0.989	0.132	1.000
CK4P-MSP	train-fit coordinate z-score	0.887	0.390	0.987
CK4+P	declared maps	0.984	0.156	1.000
CK4+P	fixed $[0,1]$ maps	0.984	0.155	1.000
CK4+P	train-fit coordinate z-score	0.869	0.404	0.840

Table S6. Complete P and MSP coordinate definitions. Population SD uses denominator L over the read-level mapped values. MSP uses bin means only in the primary representation.

block	coordinate family	definition	dimensions
P	hydrogen-bond class	Mean and population SD of A/T= 2, C/G= 3, N= 0	2
P	GC indicator	Mean and population SD of C/G= 1, A/T/N= 0	2
P	purine indicator	Mean and population SD of A/G= 1, C/T/N= 0	2
P	EIIP mapping	Mean and population SD of A= 0.1260, C= 0.1340, G= 0.0806, T= 0.1335, N= 0	2
P	ambiguity	Fraction of N bases among A/C/G/T/N characters	1
P	scaled length	Read length divided by 200	1
P	scaled entropy	Shannon entropy over A/C/G/T/N divided by $\log_2 5$	1
MSP	five per-base channels	Hydrogen, GC, purine, EIIP and N-indicator values; mean pooled within each relative-position bin	$5 \times 15 = 75$
MSP	relative-position bins	For each $b \in \{2, 3, 4, 6\}$, edges are $\lfloor jL/b \rfloor$; empty bins use a zero row; the main method excludes within-bin SD	15 bins

Table S7. Public reference assemblies used to sample the controlled WGS read templates. Each species is represented by the listed NCBI assembly accession.

genus	species	NCBI assembly accession
Acinetobacter	<i>Acinetobacter baumannii</i>	GCF.900011295.1
Acinetobacter	<i>Acinetobacter calcoaceticus</i>	GCF.000368965.1
Acinetobacter	<i>Acinetobacter nosocomialis</i>	GCF.041021905.1
Acinetobacter	<i>Acinetobacter pittii</i>	GCF.000369045.1
Burkholderia	<i>Burkholderia cenocepacia</i>	GCF.000009485.1
Burkholderia	<i>Burkholderia cepacia</i>	GCF.001411495.1
Burkholderia	<i>Burkholderia multivorans</i>	GCF.000010545.1
Candida	<i>Candida albicans</i>	GCF.000182965.3
Candida	<i>Candida dubliniensis</i>	GCF.000026945.1
Candida	<i>Candida parapsilosis</i>	GCF.000182765.1
Candida	<i>Candida tropicalis</i>	GCF.000006335.3
Enterobacter	<i>Enterobacter asburiae</i>	GCF.900077725.1
Enterobacter	<i>Enterobacter bugandensis</i>	GCF.900324475.1
Enterobacter	<i>Enterobacter cloacae</i>	GCF.905331265.2
Enterobacter	<i>Enterobacter hormaechei</i>	GCF.000213995.1
Enterobacter	<i>Enterobacter kobei</i>	GCF.023023125.1
Escherichia	<i>Escherichia coli</i>	GCF.000005845.2
Escherichia	<i>Escherichia fergusonii</i>	GCF.000026225.1
Klebsiella	<i>Klebsiella pneumoniae</i>	GCF.000240185.1
Klebsiella	<i>Klebsiella quasipneumoniae</i>	GCF.003181175.1
Klebsiella	<i>Klebsiella variicola</i>	GCF.000025465.1

Table S8. Six-target CAMI_TOY_low fixed-head summary at the prespecified block-normalized $C = 1$ probe. Clean F1 is averaged over the six 100-bp clean target tasks. Shifted F1 and retention are first averaged over eight non-baseline length-by-condition cells within each target and then over six targets. Minimum values are minima of the corresponding target-level means, not individual read-level observations. Each task used 1,200 target and 1,200 stratified background source groups with fivefold source-grouped cross-validation. The six tasks share one source pool, so cross-target summaries quantify task-level consistency.

representation	dimensions	clean F1	shifted F1	minimum target F1	retention	minimum target retention
CK4	136	0.823	0.794	0.693	0.963	0.925
CK4+P	147	0.818	0.794	0.698	0.970	0.936
CK4+MSP	211	0.818	0.793	0.698	0.969	0.933
CK4P-MSP	222	0.816	0.793	0.699	0.971	0.940
CK5	512	0.817	0.795	0.698	0.972	0.953
PseKNC	67	0.803	0.783	0.684	0.974	0.956
PseEIIP	64	0.808	0.781	0.678	0.965	0.931
NCP+ANF	400	0.692	0.661	0.530	0.953	0.897
Hashed $k = 15$	222	0.505	0.505	0.499	1.000	0.979

Table S9. Metadata for the six prespecified CAML_TOY_low target definitions. Scientific names and ranks were retrieved from NCBI Taxonomy by TaxID. Selection rank reflects abundance among eligible labels after deterministic mate collapse and exclusion of sequences shorter than 100 bp. Each one-versus-background task used 1,200 target and 1,200 stratified background source groups. The tasks share one source pool, and a source read can contribute to the background class of more than one task.

selection rank	NCBI TaxID	scientific name	taxonomic rank	target/background groups
1	434085	<i>gamma proteobacterium IMCC2047</i>	species	1,200/1,200
2	247639	<i>marine gamma proteobacterium HTCC2080</i>	species	1,200/1,200
3	1050222	<i>Paenibacillus sp. Aloe-11</i>	species	1,200/1,200
4	552396	<i>Erysipelotrichaceae bacterium 5_2_54FAA</i>	species	1,200/1,200
5	1122939	<i>Patulibacter americanus DSM 16676</i>	strain	1,200/1,200
6	1097667	<i>Patulibacter medicamentivorans</i>	species	1,200/1,200

Table S10. Coordinate-family attribution audit for the global P summary. P-property contains the eight means and population standard deviations of the hydrogen-bond, GC, purine and EIIP maps. P-auxiliary contains N fraction, length divided by 200 and normalized Shannon entropy. Global drift is the mean over 24 WGS length-by-perturbation cells; local readouts are mean macro-F1 over the 12 template-grouped spatial-localization or substitution-chemistry cells. Values are descriptive split variants of the declared full P block and are not separate model-selection results.

representation	P subset	dimension	global drift	chemistry macro-F1	spatial macro-F1
P	property moments	8	0.0232	0.6469	0.4923
P	auxiliary coordinates	3	0.0179	0.5055	0.4982
CK4+P	property moments	144	0.1556	0.6136	0.9095
CK4+P	auxiliary coordinates	139	0.1549	0.5007	0.9098
CK4P-MSP	property moments	219	0.1282	0.7104	0.9761
CK4P-MSP	auxiliary coordinates	214	0.1276	0.6868	0.9750

Table S11. Unified feature-extraction runtime benchmark. All routes received the same seeded 10,000-read, 75-bp batch after an untimed 100-read warm-up. Calls were scheduled in randomized round-robin order until every method accumulated at least 120 timed seconds and five calls. The table reports full-run medians and interquartile ranges. The late/early column is the second-half-to-first-half median ratio and exposes workstation-state drift; it is not a cross-method performance metric. The 100,000-read column is a separate single scaling pass. Disk input/output was excluded, garbage collection preceded each call and numerical libraries were restricted to one thread.

representation	features	calls	timed s	10k median [IQR], s	late/early	100k pass, s
CK4	136	77	120.9	1.29 [0.93, 2.17]	2.35	9.19
CK4+P	147	67	122.3	1.27 [1.12, 2.72]	2.41	11.66
CK4P-MSP	222	63	122.5	1.35 [1.26, 2.94]	2.34	13.09
CK4P-MSP-PKM	297	47	123.6	2.10 [2.03, 2.39]	1.08	21.16
Hashed k=15, d=222	222	40	120.7	2.87 [2.80, 2.98]	1.01	30.19
MinHash k=15, s=222	222	40	124.0	2.93 [2.89, 3.16]	0.99	30.98
CK15 random projection, d=222	222	26	125.0	4.75 [4.67, 4.91]	0.98	44.49
PseKNC (k=3, λ=3)	67	5	124.7	25.14 [24.32, 25.41]	0.96	246.18
NCP+ANF	300	128	120.7	1.05 [0.66, 1.11]	1.66	6.01
PseEIIP	64	157	120.4	0.84 [0.76, 0.88]	1.21	4.38